PRABIN POUDEL



AI / ML & AI AGENT SECURITY

Computer Engineering Student | Open to internships

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Computer Engineering student focused on applied AI, model training, and secure AI-agent systems. Build on-device inference pipelines and independent authorization workflows, supported by experience in offline-first applications and backend engineering.

SELECTED AI & SECURITY PROJECTS

Sentinel AI / SentinelOS | AI agent authorization

- Built a local-first Rust/Cedar gateway for supported agent tools, enforcing signed task contracts, default-deny policies, authenticated one-time approvals, revocation, and SQLite audit logging.
- Integrated a Python SDK and restricted Wasmtime execution; validated Rust/Python regressions across macOS, Ubuntu, and Windows CI. Research release candidate with explicit security boundaries.

Rescuer Sathi | On-device AI & offline-first systems

- Trained three models for injury detection, injury classification, and burn-severity classification; integrated TensorFlow Lite inference into Flutter. Experimental screening, not a clinical diagnosis.
- Implemented encrypted SOS persistence, retry delivery, and Nearby relay; tested relay and power behavior on physical devices. Connected responder workflows to Spring Boot and PostgreSQL.

Chitwan Care | Computer vision & environmental reporting

- Integrated a bundled five-class LiteRT waste detector with image capture, bounding boxes, and GPS reports; tested on Android with representative local images.
- Built report review, department assignment, and cleanup-evidence workflows using a deployed Spring Boot backend and Neon PostgreSQL.

Additional projects

Chitwan Plus - React/TypeScript travel planning with Gemini-assisted itineraries.
Smart University Management System - Java/Spring Boot campus workflows and PDF generation.

TECHNICAL SKILLS

AI / ML: Python, model training, YOLO / Ultralytics, TensorFlow Lite / LiteRT, computer vision
Security & systems: Rust, Cedar Policy, WebAssembly, Wasmtime, authorization, audit logging
Application engineering: Java, Dart, JavaScript, TypeScript, C/C++, Flutter, React, Spring Boot, REST APIs, JWT, Spring Data JPA, Riverpod, HTML/CSS, Vite, Leaflet
Data & tools: PostgreSQL, SQLite, Cloudflare Workers/D1, Git, Docker, GitHub Actions, Playwright

EXPERIENCE

Technology Content Writing Intern | Routine of Nepal Banda (Tech)
June 2025 - November 2025

Researched and edited technology articles; collaborated with the editorial team on accuracy and clarity.

EDUCATION

Bachelor in Computer Engineering | Ongoing
United Technical College, Pokhara University - Bharatpur, Chitwan

High School (10+2) | 2022 - 2024 | GPA: 3.7
Polar Star English Boarding School - Bharatpur, Chitwan

Languages: Nepali (Native) | English (Fluent)